



ITM 31013

OSI Model

Network Reference Models

- All networks provide the same basic functions to transfer a message from sender to receiver, but each network can use different network hardware and software to provide these functions.
- All of these hardware and software products have to work together to successfully transfer a message.

Network Reference Models

- When the networks came into being, computers were typically was able to communicate only with a device from same manufacturer.
- So there was a necessity to create standards to ensure the vendors can create interoperable network implementations including hardware.
- So few different organizations created these standards using Network Reference Models.

Network Reference Models

- A **reference model** is a conceptual blueprint of how communications should take place. it addresses all the processes required for effective communication and divides these processes into logical grouping called layers.
- The most important reference models are :
 1. OSI reference model.
 2. TCP/IP reference model OR TCP/IP Protocol suite.

OSI reference model

- The Open Systems Interconnection (OSI) reference model is used to define how data communication occurs between devices.
- Open Systems Interconnection (OSI) reference model was created by ISO in late 1970s .
- The model is divided into 7 layers.
- The **purpose of each layer** in the OSI reference model is to provide service to the next layer above it while shielding the upper layer from the complications of the layer below it.

Functions of OSI Model

- Helps 'Break down' network functions.
- Creates standards for equipment manufacturing.
- Allows vendors to focus in specialized areas of the network.

Purpose of the OSI Model

The OSI model serves the following purposes:

Standardization – It provides a universal standard for different networking systems and devices to communicate.

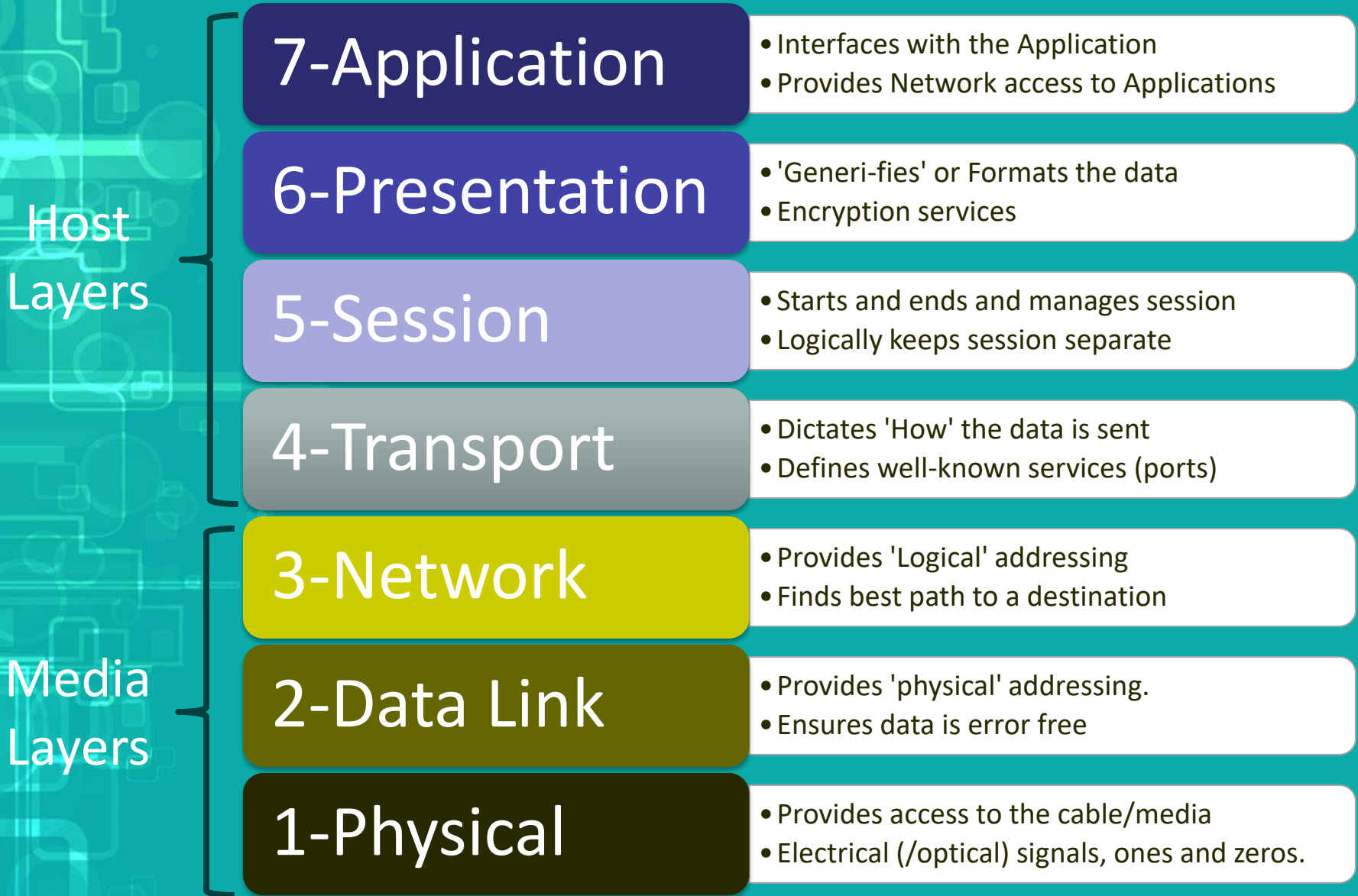
Interoperability – Different hardware and software from various vendors can work together seamlessly.

Modularity – It breaks down networking functions into separate, manageable layers, making it easier to troubleshoot and upgrade.

Error Handling & Security – Different layers manage error detection, correction, and data security.

Encapsulation & Abstraction – It ensures that each layer handles specific tasks without worrying about the complexities of other layers.

7 Layers of OSI Reference Model



OSI Model Layers

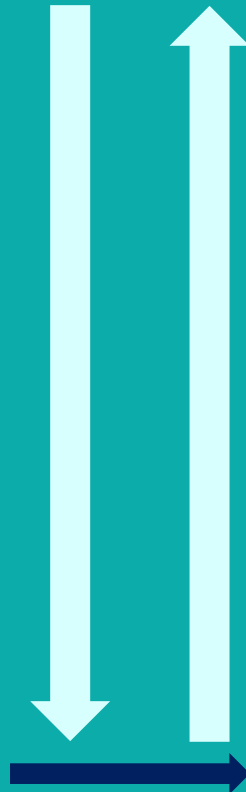


Sender



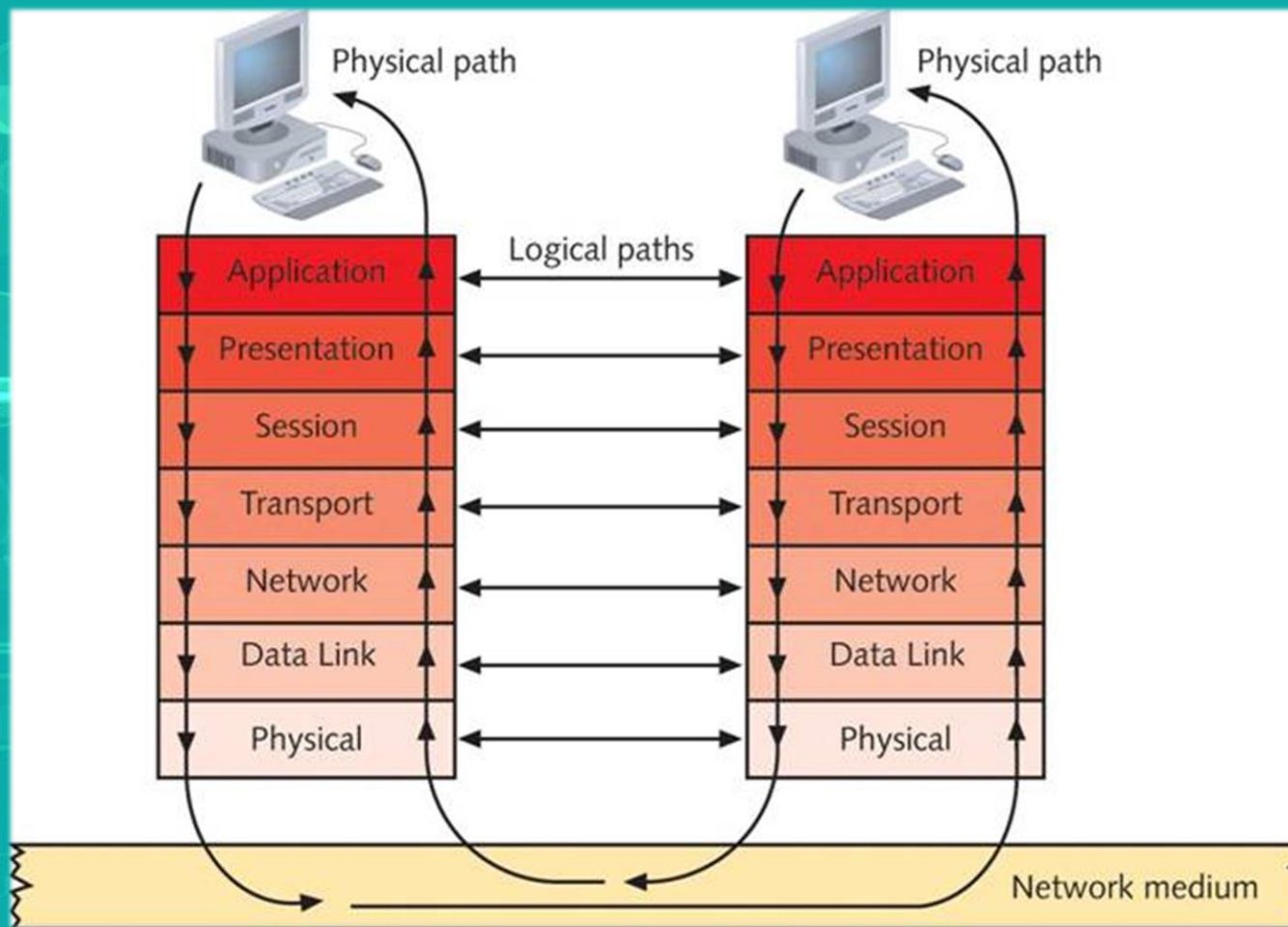
Receiver

Layer	Protocol Data Unit (PDU)
Application	Data
Presentation	Data
Session	Data
Transport	Segment
Network	Packet
Data Link	Frame
Physical	Bits



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The OSI Model



Flow of data through the OSI Model

The OSI Model

On sending computer

- Sending application access the application layer.
- Application provides data to the presentation layer.
- Presentation layer format the data as per network requirement and forward it's to session layer.
- Session layer initiate the connection and forward the data to the transport layer.
- Transport layer broke down the large data file in smaller segments and add a header with control information, which are bits designated to describe how to determine whether the data is complete, uncorrupted, in the correct sequence, and so forth.

The OSI Model

On receiving computer

- Physical layer receive signals from media and convert them in frames. Frames are forwarded to the data link layer.
- Data link layer check the frame. All tampered frame are dropped here. If frame is correct, data link layer strip down its header and footer from frame and hand over packet to network layer.
- Network layer check the packet with its own implementations. If it's found everything fine with packet, it strips down its header from packet and hand over segment to transport layer.

The OSI Model

On sending computer (Contd...)

- Segments are forwarded to the network layer. Network layer add its header, with logical address and convert it in packet. Network layer forwards packet to data link layer.
- Data link layer attach its header and footer to the packet and convert it in frame.
- Frames are forwarded to the physical layers that convert them in signals. These signals are loaded in media.

The OSI Model

On receiving computer(Contd...)

- Transport layer again do the same job. It verifies the segments with its own protocol rules. Only the verified segments are processed. Transport layer remove its header from verified segments and reassemble the segments in data. Data is handed over the session layer.
- Session layer keep track of open connection and forwarded the receiving data to presentation layer.
- Presentation form the data in such a way that application layer use it.
- Application layer on receiving computer find the appropriate application from the computer and open data within particular application.

Application Layer

- Application layer marks the spot where users actually communicate with the computer/host.
- All applications and utilities that communicate with network fall in this layer (some are directly and some are indirectly).
- Example Applications
 - **Browsers** :- Mozilla Firefox, Internet Explorer, Google Chrome etc
 - **Email clients**: - Microsoft Outlook, Mozilla Thunderbird etc.
 - **FTP clients** :- Filezilla

Application Layer

Responsibilities:

- Provides interface to the users to send/receive data over the network.
- Determines the source and destination of the intended communication.
- Determine whether the sufficient resources exist for the intended communication.
- Determines authorization and authentication of communicating end users
- Provides services for uniting the components of network applications, such as remote access, directory access, e-mail, and file transfer.

Application Layer

Protocols used in this layer:

- **HTTP (*Hypertext Transfer Protocol*)** — Used to browse web pages.
- **POP3 (*Post Office Protocol v.3*)** — Used to retrieves email.
- **SMTP (*Simple Mail Transfer Protocol*)** — Used to sends email.
- **FTP (*File Transfer Protocol*)** — Used to reliably sends/retrieves files.
- **TFTP (*Trivial File Transfer Protocol*)** — Used to transfer the files rapidly.
- **Telnet** — used to connect remote devices.
- **SNMP (*Simple Network Management Protocol*)** — Used to control the connected networking devices.
- **DNS (*Domain Naming System*)** — Used to translate the name with IP address and vice versa.
- **DHCP (*Dynamic Host Configuration Protocol*)** — Used to assign IP address and DNS information automatically to hosts.
- **NTP (*Network Time Protocol*)** — Used to synchronizes clocks.

Presentation Layer

- The presentation layer formats the data to be presented to the application layer.
- It can be viewed as the translator for the network.
- This layer may translate data from a format used by the application layer into a common format at the sending station, then translate the common format to a format known to the application layer at the receiving station.

Presentation Layer

- Responsibilities:
 - Translates different application formats into standard format, and vice versa
 - Transforms data with different structures into a standard format
 - Formats data transfer syntax
 - Performs encryption and decryption of data before transmitting across the network
 - Performs data compression and decompression while transmitting data
 - Performs conversion between graphic standards

Presentation Layer

Protocols used in this layer:

- TLS - Transport Layer Security

Its predecessor is called Secure Sockets Layer (SSL). Both protocols frequently referred to as "SSL". These are cryptographic protocols that provide communications security over a computer network.

- AFP - Apple Filing Protocol

This is proprietary network protocol used by Apple to provide file services for Mac Operating system.

Session Layer

- Session layer deals with connections. It establishes, manages, and terminates sessions between two communicating nodes.

Responsibilities:

- Establishes, maintains, and terminates the connection between presentation layer entities.
- Manages more than one connection for one application
- Coordinates the interaction between two applications
- Reconnects two hosts in case of an error
- Coordinates dialogs and data exchange
- Determines the duration of a session

Transport Layer

- Protocols accept data from Session layer and manage end-to-end delivery of data
 - Ensure data transferred reliably, in correct sequence, and without errors
- Protocols also handle flow control
 - Determine the appropriate rate of transmission based on how fast recipient can accept data

Transport Layer (continued)

- Transmission Control Protocol (TCP): Takes care of reliably transmitting HTTP requests from client to server and vice versa
- Connection-oriented protocols: ensure that data arrives exactly as it was sent
 - Establish connection before transmitting data
 - TCP is connection-oriented
 - Client's TCP protocol first sends synchronization (SYN) packet request to server
 - Server responds with synchronization-acknowledgment (SYN-ACK) packet
 - Client responds with own acknowledgment (ACK)

Transport Layer (continued)

- Acknowledgments also used to ensure that data was properly delivered
 - For every data unit sent, connection-oriented protocol expects acknowledgment from recipient
 - If no acknowledgment, data retransmitted
- Connection-oriented protocols use a checksum
 - Unique character string allowing receiving node to determine if arriving data unit exactly matches data unit sent by source

Transport Layer (continued)

- Connectionless protocols do not establish connection before transmitting
 - No effort to ensure data delivered error-free
- Transport layer protocols break large data units received from Session layer into smaller segments (segmentation)
- Maximum transmission unit (MTU): largest data unit a given network will carry

Transport Layer (continued)

- Reassembly: process of reconstructing segmented data units
- Sequencing: method of identifying segments that belong to same group of subdivided data
 - Indicates where unit of data begins
 - Indicates order in which groups of data were issued
 - Transport layer protocols of two nodes must synchronize timing and agree on starting point for the transmission

Transport Layer (continued)

Responsibilities

- Transfers data between hosts
- Performs error checking and recovery in case of loss of transmission
- Ensures data integrity by maintaining flow control
- Provides acknowledgement for successful transmissions
- Ensures complete data transfers using techniques such as CRC, windowing, and acknowledgements

Network Layer

- Primary functions of protocols:
 - Translate network addresses into physical counterparts
 - Decide how to route data from sender to receiver
- Each node has two types of addresses:
 - Network address: follows hierarchical addressing scheme
 - Can be assigned through OS software
 - Network layer addresses, logical addresses, or virtual addresses
 - This is called IP addresses in another term.
 - IP address is used in Network Layer.
 - Physical address
 - This address is used in Data Link layer.
 - In another term this address is called as MAC address.

Network Layer (continued)

- Network layer protocols accept Transport layer segments and add logical addressing information in network header
- Network layer handles routing
 - Determining best network path
- IP: Network layer protocol that underlies most Internet traffic
- Fragmentation: Network layer protocol subdivides segments it receives from Transport layer into smaller packets

Network Layer (continued)

Responsibilities

- Path selection between end-systems (routing)
- Subnet flow control.
- Fragmentation & reassembly
- Translation between different network types

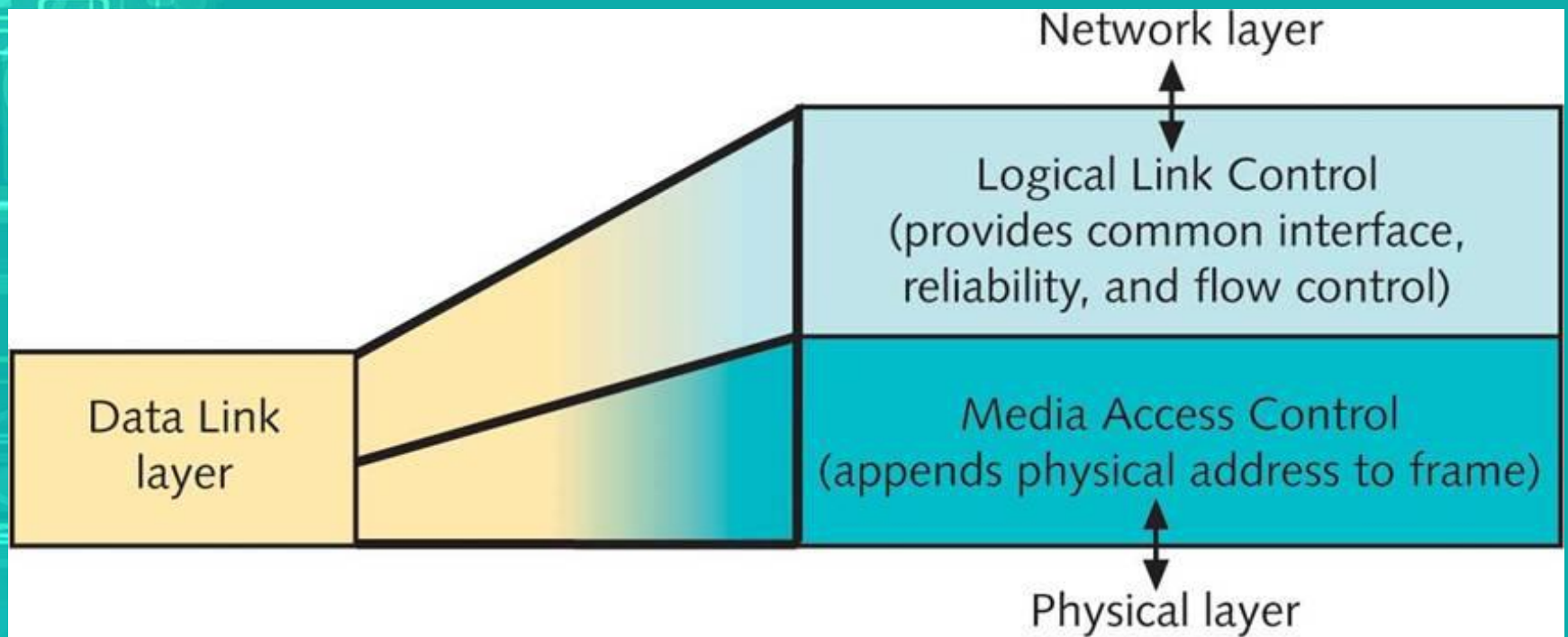
Data Link Layer

- Protocols divide received data into distinct frames
 - Can then be transmitted by Physical layer
- Frame: structured package for moving data
 - Raw data
 - “payload”
 - Sender’s and receiver’s network addresses
 - Error checking and control information

Data Link Layer (continued)

- Error checking accomplished by 4-byte Frame Check Sequence (FCS) field
 - Ensures data at destination exactly matches data issued from source
 - When source node transmits data, performs Cyclic Redundancy Check (CRC) to get FCS
 - Destination node's Data Link layer services unscramble FCS via same CRC algorithm
- Data Link layer divided into two sub-layers:
 - Logical Link Control
 - Media Access Control

Data Link Layer (continued)



The Data Link layer and its sublayers

Data Link Layer (continued)

- Logical Link Control (LLC) sublayer:
 - Provides interface to Network layer protocols
 - Manages flow control
 - Issues requests for transmission for data that has suffered errors
- Media Access Control (MAC) sublayer:
 - Manages access to physical layer
 - Appends destination computer's physical address onto data frame (MAC address, Data Link layer address, or hardware address)

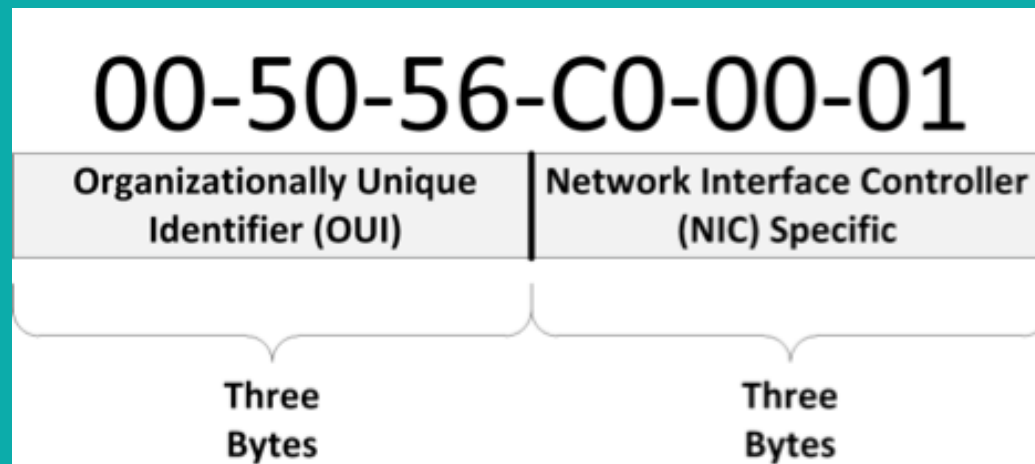
Data Link Layer (continued)



A NIC's MAC address

Data Link Layer (continued)

- MAC addresses contain two parts:
 - Block ID: six-character sequence unique to vendor
 - Device ID: six-character sequence based on NIC's model and manufacture date



Physical Layer

- Protocols accept frames from Data Link layer and generate voltage to transmit signals
- When receiving data, protocols detect voltage and accept signals
- Protocols also set data transmission rate and monitor data error rates
 - Cannot perform error correction
- NICs operate at both Physical layer and Data Link layer
- Network administrators mostly concerned with bottom four layers of OSI Model

Applying the OSI Model

OSI Model Layer	Function
Application (Layer 7)	Provides interface between software applications and network for interpreting applications' requests and requirements
Presentation (Layer 6)	Allows hosts and applications to use a common language; performs data formatting, encryption, and compression
Session (Layer 5)	Establishes, maintains, and terminates user connections
Transport (Layer 4)	Ensures accurate delivery of data through flow control, segmentation and reassembly, error correction, and acknowledgment
Network (Layer 3)	Establishes network connections; translates network addresses into their physical counterparts and determines routing
Data Link (Layer 2)	Packages data in frames appropriate to network transmission method
Physical (Layer 1)	Manages signaling to and from physical network connections

Functions of the OSI layers



Data representation and encryption

Recognizing data: HTML, DOC, JPEG, MP3, AVI, Sockets

6. Presentation



Interhost communication

Session establishment in TCP, SIP, RTP, RPC-Named pipes

5. Session



End-to-end connections and reliability

TCP, UDP, SCTP, SSL, TLS

4. Transport



Path determination and logical addressing

IP, ARP, IPsec, ICMP, IGMP, OSPF

3. Network

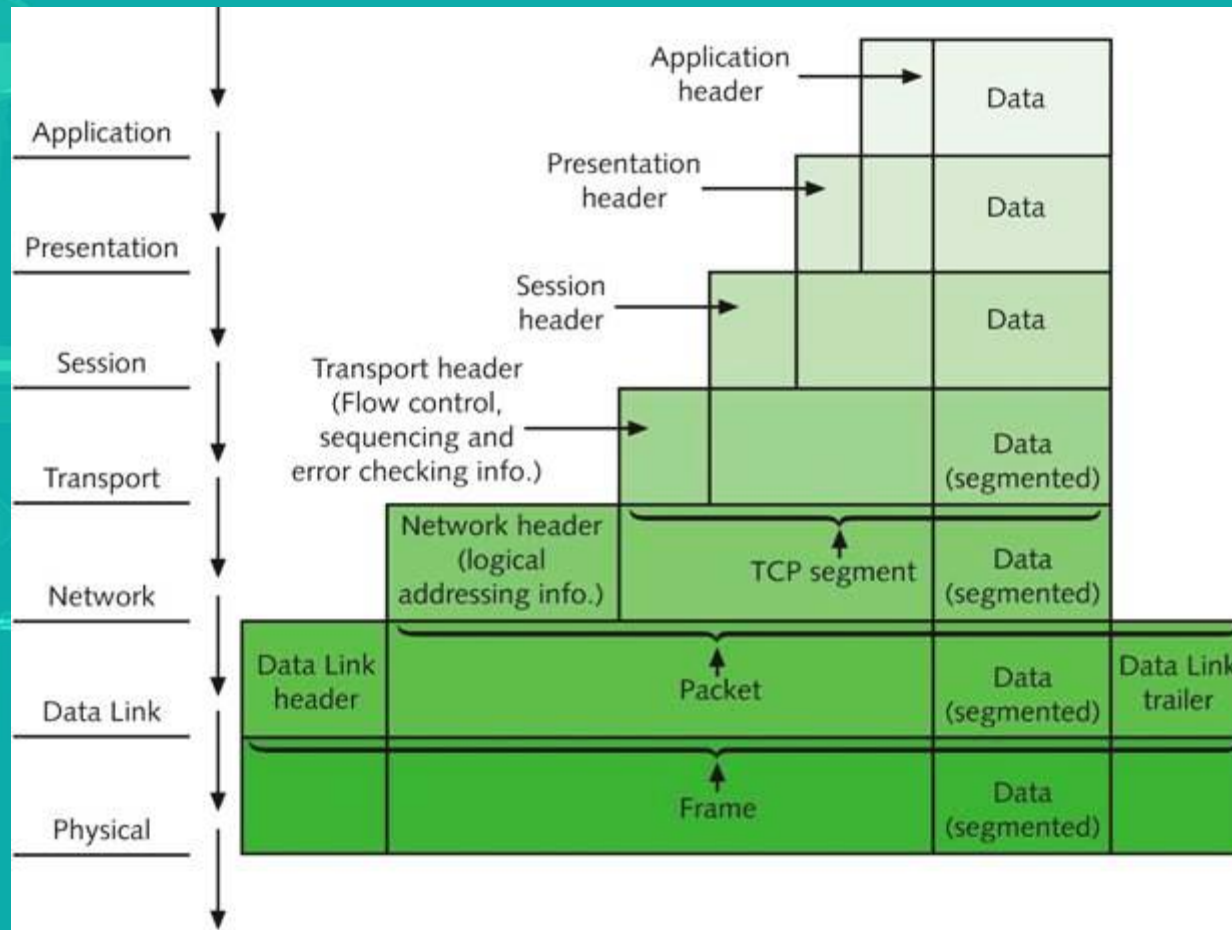


Physical addressing

Ethernet, 802.11, MAC/LLC, VALN, ATM, HDP, Fibre Channel,

2. Data Link

OSI Model Encapsulation



Data transformation through the OSI Model

OSI Model Encapsulation (Contd..)

OSI Model - Encapsulation

OSI Layer	Wrapper Name	Header Name	Frequency of usage
Application	N/A	Layer 7 Header	<i>Everytime (100%)</i>
Presentation	N/A	Layer 6 Header	<i>Everytime (100%)</i>
Session	N/A	Layer 5 Header	<i>Everytime (100%)</i>
Transport	Segment	TCP Header	<i>More common</i>
		UDP Header	<i>Less common</i>
Network	Packet	IPv4 Header	<i>Old Standard, still more common</i>
		IPv6 Header	<i>New Standard, still less common</i>
Data Link	Frame	Ethernet Type II Header	<i>More common</i>
		IEEE 802.2 802.3 802.3 SNAP	<i>Less common</i>
		Other Frame Headers	
Physical	Bits	N/A	<i>Everytime (100%)</i>

Original - JonathanCrosby.com

OSI Model compared to TCP Model

OSI Model	TCP Model
Application Layer	Application Layer
Presentation Layer	
Session Layer	
Transport Layer	Transport Layer
Network Layer	Internet Layer
Data Link Layer	Network Access Layer
Physical Layer	

Frame Specifications

- The two major categories of frame types:
 - Ethernet
 - Four types of Ethernet frames
 - Most popular form characterized by unique way in which devices share a common transmission channel (described in IEEE 802.3 standard)
 - Token Ring: relies on direct links between nodes and a ring topology
 - Nodes pass around tokens (control frames that indicate to network when a node is about to transmit data)

Common Network Port Numbers

- Ports allow a single host with a single IP address to run network services.
- Each port number identifies a distinct service, and each host can have 65535 ports per IP address.
- Port use is regulated by the Internet Corporation for Assigning Names and Numbers (ICANN).

Common Network Port Numbers

- By ICANN there are three categories for ports:
 - From 0 to 1023 – well known ports assigned to common protocols and services
 - From 1024 to 49151 – registered ports assigned by ICANN to a specific service
 - From 49152 to 65535 – dynamic (private, high) ports range from 49,152 to 65,535. Can be used by any service on an ad hoc basis. Ports are assigned when a session is established, and released when the session ends.

Common Network Port Numbers

Port	Service name	Transport protocol
20, 21	File Transfer Protocol (FTP)	TCP
22	Secure Shell (SSH)	TCP and UDP
23	Telnet	TCP
25	Simple Mail Transfer Protocol (SMTP)	TCP
50, 51	IPSec	
53	Domain Name Server (DNS)	TCP and UDP
67, 68	Dynamic Host Configuration Protocol (DHCP)	UDP
69	Trivial File Transfer Protocol (TFTP)	UDP
80	HyperText Transfer Protocol (HTTP)	TCP

Common Network Port Numbers

Port	Service name	Transport protocol
80	HyperText Transfer Protocol (HTTP)	TCP
110	Post Office Protocol (POP3)	TCP
119	Network News Transport Protocol (NNTP)	TCP
123	Network Time Protocol (NTP)	UDP
135-139	NetBIOS	TCP and UDP
143	Internet Message Access Protocol (IMAP4)	TCP and UDP
161, 162	Simple Network Management Protocol (SNMP)	TCP and UDP
389	Lightweight Directory Access Protocol	TCP and UDP
443	HTTP with Secure Sockets Layer (SSL)	TCP and UDP